COMPATIBILITY OF THE GYSIN MAP WITH A FLAT PULLBACK OF THE AMBIENT SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xypic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Flat pullback of cycle classes of coherent sheaves and compatibility of the cap with a first Chern class and flat pullback. Same chapter chow.tex, labels lemma-pullback-coherent (proof line 5328, claimed at a withdrawn revision and re-established) and lemma-flat-pullback-cap-c1 (proof line 5331, claimed as S14-algebra-flat-pullback-cap-c1); the pullback diagram is described in remark-pullback-pairs (proof line 5314). The first handles the case where the component is not contained in the divisor, identifying the two intersections as inverse images, and the second handles the case where it is, where the equality is the compatibility of the cap with flat pullback.
- Infinite sums of rational equivalences. Same chapter chow.tex, label remark-infinite-sums-rational-equivalences (proof line 5332). Extends the equality from a single integral class to a general cycle.

Source reference. Original definition 02T8: sources/stacks-chow/pinned-original-source.tex, source lines 5069–5069. Navigation pointer for \ref{definition-gysin-homomorphism} used inside the counted span (line 5307); the argument that consumes it is retained in the span.

Source reference. Original lemma 02SS: sources/stacks-chow/pinned-original-source.tex, source lines 4533–4533. Navigation pointer for \ref{lemma-flat-pullback-cap-c1} used inside the counted span (line 5331); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02RE: sources/stacks-chow/pinned-original-source.tex, source lines 2331–2331. Navigation pointer for \ref{lemma-pullback-coherent} used inside the counted span (line 5328); the argument that consumes it is retained in the span.

Source reference. Original remark 02RZ: sources/stacks-chow/pinned-original-source.tex, source lines 3057–3057. Navigation pointer for \ref{remark-infinite-sums-rational-equivalences} used inside the counted span (line 5332); the argument that consumes it is retained in the span.

Source reference. Original remark 0B6Y: sources/stacks-chow/pinned-original-source.tex, source lines 5248–5248. Navigation pointer for \ref{remark-pullback-pairs} used inside the counted span (line 5314); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 5304); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let $f : X' \to X$ be a flat morphism of relative dimension r of schemes locally of finite type over S. Let $(\mathcal{L}, s, i : D \to X)$ be as in Definition 02T8. Form the diagram*

$$\begin{array}{ccc} D' & \xrightarrow{i'} & X' \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ D & \xrightarrow{i} & X \end{array}$$

as in Remark 0B6Y. For any $(k+1)$-cycle α on X we have $(i')^* f^* \alpha = g^* i^* \alpha$ in $\mathrm{CH}_{k+r}(D')$ (this makes sense as f^* is defined on the level of cycles).

Proof. Suppose $\alpha = [W]$ for some integral closed subscheme $W \subset X$. Let $W' = f^{-1}(W) \subset X'$. In case $W \not\subset D$, then $W' \not\subset D'$ and we see that

$$W' \cap D' = g^{-1}(W \cap D)$$

as closed subschemes of D'. Hence the equality holds as cycles, see Lemma 02RE. In case $W \subset D$, then $W' \subset D'$ and $W' = g^{-1}(W)$ with $[W']_{k+1+r} = g^*[W]$ and equality holds in $\mathrm{CH}_{k+r}(D')$ by Lemma 02SS. By Remark 02RZ the result follows for general α'. $\square$